Unit 20: Welfare Effects of Monopoly

Objectives of the Study:

After studying this unit, the student should be able to

-know the welfare loss as a result of monopoly practices in the economy.

A monopoly is a market structure in which a single firm is the sole producer of a good or service with no close substitutes. Unlike perfectly competitive firms, a monopolist has market power, allowing it to influence the price of its product. While monopolies can earn high profits, their pricing and output decisions have important implications for economic welfare, particularly for consumers and society as a whole.

Meaning of Economic Welfare

Economic welfare refers to the overall well-being and satisfaction of individuals participating in a market, including both consumers and producers. It is commonly measured using two key concepts: consumer surplus and producer surplus. Consumer surplus (CS) represents the benefit that consumers receive when they are willing to pay more for a good than the actual price they pay in the market. In other words, it reflects the extra satisfaction or value gained by consumers. Producer surplus (PS), on the other hand, refers to the benefit received by producers when the price they receive for a product is higher than the minimum cost of producing it. It represents the earnings or gains to producers above their production costs.

The sum of consumer surplus and producer surplus gives total economic welfare, which indicates the overall efficiency and well-being generated in a market:

$$\text{Total Welfare} = \text{Consumer Surplus} + \text{Producer Surplus}$$

A higher total welfare implies that resources are being allocated efficiently and that both consumers and producers are benefiting from market transactions. Economists use this concept to compare different market structures, such as perfect competition and monopoly, and to evaluate how policies or market power affect the well-being of society.

Welfare Effects of Monopoly

The welfare effects of monopoly refer to how a monopoly market structure impacts the overall well-being of society, including both consumers and producers. In a monopoly, a single firm controls the market and has the power to set prices above marginal cost, unlike in perfect competition where price equals marginal cost.

As a result, a monopoly produces less output and charges a higher price, which leads to several important welfare consequences:

1. Reduction in Consumer Surplus

Under a monopoly, the firm has the power to control the market price, and it typically sets a higher price than would exist in a competitive market. At this higher price, many consumers either reduce their consumption or are unable to afford the product altogether. As a result, the quantity demanded decreases, meaning fewer units of the good are sold in the market.

This situation leads to a significant reduction in consumer surplus, which represents the difference between what consumers are willing to pay and what they actually pay. Because the price is higher, consumers pay more for each unit they purchase, reducing the benefit they receive. Additionally, some consumers who would have bought the product at a lower price are now excluded from the market, as they are not willing or able to pay the higher monopoly price.

Thus, monopoly pricing not only reduces the satisfaction of existing consumers but also eliminates potential gains for those who are priced out, resulting in a substantial loss of consumer welfare.

2. Increase in Producer Surplus (Profit)

In a monopoly, the firm has the power to set prices above marginal cost, which allows it to earn supernormal (abnormal) profits. By restricting output and charging a higher price, the monopolist increases its total revenue relative to its costs. As a result, the firm captures a larger producer surplus, which represents the difference between the price it receives and the cost of production.

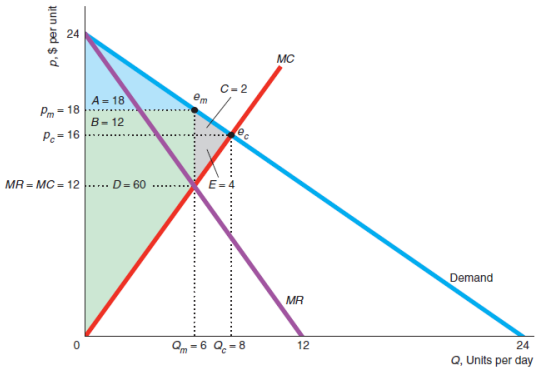
Part of the consumer surplus that is lost due to higher prices is effectively transferred to the producer in the form of higher profits. Consumers who continue to purchase the product now pay more than they would in a competitive market, and this extra payment becomes additional income for the monopolist.

However, this gain in producer surplus does not fully compensate for the loss in consumer welfare. This is because some consumers are completely priced out of the market and no longer purchase the product. The value that these consumers would have gained is lost entirely and is not captured by the producer. This creates a net loss in total welfare, indicating that monopoly leads to inefficiency despite the increase in producer profits.

3. Deadweight Loss (Market Failure Due to Monopoly Pricing)

Unlike perfect competition, which achieves economic efficiency by maximizing welfare, W (= consumer surplus + producer surplus = CS + PS), a profit-maximizing monopoly is economically inefficient because it wastes potential surplus, resulting in a deadweight loss, DWL. The inefficiency of monopoly pricing is an example of a market failure: inefficient production or consumption, often because a price exceeds marginal cost. The market failure from a monopoly occurs because its price is greater than its marginal cost. This economic inefficiency creates a rationale for governments to intervene.

Figure 19.3: Deadweight Loss



We illustrate this loss using our continuing example. If the monopoly were to act like a competitive market and operate where its inverse demand curve, intersects its marginal cost (supply) curve,

$$p = 24 - Q = 2Q = MC,$$

it would sell $Q_c = 8$ units of output at a price of 16, as in Figure 19.3. At this competitive price, consumer surplus is area A + B + C and producer surplus is D + E. If the firm acts like a monopoly and operates where its marginal revenue equals its marginal cost, only 6 units are sold at the monopoly price of 18, and consumer surplus is only A. Part of the lost consumer surplus, B, goes to the monopoly, but the rest, C, is lost.

By charging the monopoly price of 18 instead of the competitive price of 16, the monopoly receives 2 more per unit and earns an extra profit of area B = 12 on the Qm = 6 units it sells. The monopoly loses area E, however, because it sells less than the competitive output. Consequently, the monopoly's producer surplus increases by B - E over the competitive level. We know that its producer surplus increases, B - E >0, because the monopoly had the option of producing at the competitive level and chose not to do so.

Welfare under monopoly is lower than with competition. The deadweight loss of monopoly is -C - E, which is the consumer surplus and producer surplus that is lost because less than the competitive output is produced. The deadweight loss is due to the gap between price and marginal cost at the monopoly output. At Qm = 6, the price, 18, is above the marginal cost, 12, so consumers are willing to pay more for the last unit of output than it costs to produce it. We can summarise this into the table 19.1.

Table 19.1: Deadweight Loss

	Competition	**Monopoly**	**Change**
Consumer Surplus, CS	A + B + C	A	-B-C= ΔCS
Producer Surplus, PS	D +E	B+D	B-E= ΔPS
Welfare, W = CS + PS	A + B + D+E	A+B+D	-C-E=ΔW=DWL

4. Allocative Inefficiency

Allocative inefficiency is one of the major welfare drawbacks of a monopoly. In a monopolistic market, the firm sets its price above marginal cost (P > MC) in order to maximize profit. This creates a situation where the price consumers are willing to pay for the good is higher than the cost of producing it. In other words, the value that consumers place on additional units of the good exceeds the resources required to produce those units.

However, despite this potential benefit, the monopolist deliberately restricts output to keep prices high and maximize profits. As a result, fewer goods are produced and sold than would

be socially optimal. This means that some mutually beneficial transactions—where consumers value the product more than its production cost—do not take place.

Because of this underproduction, resources are not allocated efficiently in the economy. The economy fails to produce the level of output that would maximize total welfare. In contrast, in a perfectly competitive market, price equals marginal cost (P = MC), ensuring that resources are allocated in the most efficient way. Thus, monopoly leads to allocative inefficiency by limiting output and preventing the full realization of potential gains from trade.

5. Productive Inefficiency

Productive inefficiency is another possible welfare effect of monopoly that arises due to the absence of competition. In a competitive market, firms face constant pressure from rivals to minimize costs and improve efficiency in order to survive. However, a monopolist, being the sole producer, does not face such competitive pressure, which can reduce its incentive to operate efficiently.

As a result, monopolies may experience higher production costs, since they are less motivated to use resources in the most cost-effective way. They may also show less innovation, as there is no immediate threat from competitors introducing better or cheaper products. Additionally, monopolies may suffer from managerial inefficiency, where managers do not exert maximum effort or fail to control costs effectively due to the lack of accountability and competition.

This situation is often referred to as “X-inefficiency,” where the firm operates at a higher cost than necessary. Therefore, even though monopolies can earn high profits, they may not produce goods at the lowest possible cost, leading to inefficiency and a waste of economic resources.

6. Income Inequality

Monopoly can contribute to income inequality because the benefits of market power are unevenly distributed between producers and consumers. In a monopolistic market, the firm earns supernormal profits, which are often concentrated in the hands of a few owners, shareholders, or top managers. This leads to a situation where a significant portion of income is accumulated by a small group of people.

At the same time, consumers are required to pay higher prices for the goods or services provided by the monopoly. This reduces their purchasing power and overall welfare, especially for lower-income groups who may be more affected by price increases. Some consumers may even be priced out of the market entirely, further worsening their economic condition.

As a result, monopoly power can widen income inequality in society, as wealth becomes concentrated among producers while consumers bear higher costs. This unequal distribution of economic benefits highlights one of the broader social concerns associated with monopolistic markets.

The welfare effects of monopoly show that while monopolies can generate high profits for producers, they do so at the expense of consumers and overall economic efficiency. By restricting output and charging higher prices, monopolies reduce consumer surplus and create deadweight loss. As a result, monopolies lead to allocative inefficiency and reduced social welfare, making them less desirable compared to perfectly competitive markets. Overall, the welfare effects of monopoly show that although the producer may gain higher profits, society as a whole experience a loss in total economic welfare, making monopoly less efficient compared to perfectly competitive markets.

Multiple Choice Questions

1. Which of the following best describes the concept of economic welfare?

   A. The overall well-being of individuals participating in a market, measured by consumer and producer surplus.

   B. The total profits earned by monopolists in a given market.

   C. The maximum output produced in a perfectly competitive market.

   D. The difference between the highest and lowest prices in a market.

Answer: A. The overall well-being of individuals participating in a market, measured by consumer and producer surplus.

2. What effect does a monopoly have on consumer surplus compared to a perfectly competitive market?

   A. Consumer surplus increases due to lower prices.

   B. Consumer surplus remains unchanged as output is maximized.

   C. Consumer surplus decreases because monopolies set higher prices.

   D. Consumer surplus is eliminated completely in monopoly markets.

   Answer: C. Consumer surplus decreases because monopolies set higher prices.

3. In a monopolistic market structure, how does a monopolist typically maximize profit?

   A. By producing goods at the marginal cost.

   B. By setting prices equal to the average total cost.

   C. By restricting output and setting prices above marginal cost.

   D. By increasing production to meet consumer demand.

   Answer: C. By restricting output and setting prices above marginal cost.

4. What is a significant consequence of productive inefficiency in monopolies?

   A. Higher quality products due to lack of competition.

   B. Lower production costs leading to increased profits.

   C. Increased costs of production due to the absence of competitive pressure.

   D. Improved innovation as firms compete for market share.

   Answer: C. Increased costs of production due to the absence of competitive pressure.

5. How does monopoly contribute to income inequality in society?

A. By evenly distributing profits among all consumers.

B. By concentrating profits among a small group of producers while consumers face higher prices.

C. By providing equal access to goods for all income levels.

D. By lowering prices for everyone, increasing purchasing power.

Answer: B. By concentrating profits among a small group of producers while consumers face higher prices.